

Ahmed Loughzali

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Availability: Seeking a 6-month research internship starting January/February 2027

Research interests : Embodied AI and world models, with applications to robotics and medical vision.

EDUCATION

Gwangju Institute of Science and Technology (GIST)

Fall 2026

Exchange Semester — Robotics & Physical AI

Gwangju, South Korea

- Relevant Coursework: 3D vision, Robot control, World models, Autonomous navigation

Institut Mines Telecom Nord Europe (Réseau IMT)

2022 – 2027

MSc in Computer Science — Major: AI & Data Science

Lille, France

- Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Embedded Systems, Signal Processing

PUBLICATIONS (*Under review at NeurIPS 2026 workshops*)

- *Adversarial Attacks Expose Path-Dependent Failures in Discrete Medical Image Tokenizers*, first author (AlgoVerse fellowship research program)
- *Multi-Party Backchannel Prediction: a Diagnosis, a Benchmark, and a Ceiling*, second author (*work conducted at Enchanted Tools*)

PROFESSIONAL EXPERIENCE

Enchanted Tools

April 2026 – August 2026

AI & HRI Research intern

Paris, France

- Brought real-time turn-taking, user barge-in, and backchanneling capabilities to Mirokaï, a social humanoid robot, enabling natural spoken interaction (paper under review).
- Built an end-to-end multilingual barge-in detection system (12 languages): synthetic data generation, model training, and real-time deployment on Jetson Orin NX under strict latency constraints (quantization, cross-compilation, gRPC, GitLab CI/CD).
- Benchmarked state-of-the-art turn-taking models (VAP, TurnGPT, SmartTurn) and fine-tuned Qwen-based models for multilingual end-of-turn detection.
- Selected to present the robot to enterprise clients at VivaTech and Santexpo (Microsoft partnership).

Ferromobile

May 2025 – August 2025

AI Engineer intern

Paris, France

- Built and optimized a real-time Text-to-Speech service (FastAPI, Piper TTS) on NVIDIA Jetson Xavier, achieving ~0.6s latency to deliver voice guidance to passengers of an autonomous vehicle.
- Designed and deployed a local RAG pipeline (LangChain, ChromaDB, Mistral 7B) with a web interface for intelligent querying of technical documents.
- Containerized the full architecture (Docker Compose, Dev Containers) for reproducible deployment on embedded industrial hardware.

SELECTED PROJECTS

World models Hackathon (endorsed by Yann LeCun)

June 2026

- Built an action-conditioned surgical JEPA (EB-JEPA) that predicts future latent states from wrist-camera video and proprioception, applied to surgical robotics.
- Trained the model on an HPC cluster (8× NVIDIA H100 GPUs) with SLURM-managed distributed training.

OncovIA (Mistral AI x GE Healthcare Hackathon — 1st place)

February 2026

- Multimodal oncology pipeline combining Computer Vision and Agentic RAG to help radiologists generate accurate, patient-specific follow-up reports.

SKILLS & INTERESTS

Programming & Tools: Python, C++, Bash, Git, Docker, SLURM, gRPC, FastAPI

Robotics & Simulation: ROS 2, MuJoCo, Isaac Sim/Gym

ML & Edge AI: PyTorch, Hugging Face, TensorRT, Quantization, Jetson (Orin/Xavier/Thor)

Languages: Arabic & French (Native), English (C1), Spanish (B1), Russian (A2)

Interests : Hiking, reading, wrestling